

E.G.S. PILLAY ENGINEERING COLLEGE

(Autonomous)

NAGAPATTINAM – 611002

(Affiliated to Anna University, Chennai | Accredited by NAAC with 'A++'
Grade|Accredited by NBA T1(B.E. – CSE, CIVIL, ECE, EEE, MECH& B.Tech – IT) |
Approved by AICTE, New Delhi)



B.TECH - INFORMATION TECHNOLOGY R - 2023

THIRD YEAR

CURRICULUM AND SYLLABUS FOR FIFTH SEMESTER

Course Code	Course Name	Category	L	T	P	C	Max. Marks		
							CA	ES	Total
THEORY COURSES									
2302IT501	Object Oriented Analysis and Design	PCC	3	0	0	3	40	60	100
2302IT502	Web Programming	PCC	3	0	0	3	40	60	100
2302IT503	Mobile Computing	PCC	3	0	0	3	40	60	100
2302IT504	Artificial Intelligence and Machine Learning	PCC	3	0	2	4	50	50	100
	Elective I	PEC	3	0	0	3	40	60	100
	Open Elective I	OEC	3	0	0	3	40	60	100
LABORATORY COURSES									
2302IT551	Web Programming Lab	PCC	0	0	2	1	60	40	100
2304IT551	CASE Tools Lab (Mini Project I)	EEC	0	0	2	1	60	40	100
	Professional Development Course III	EEC	0	0	2	1	60	40	100
2304IT552	Startup Opportunities for IT Engineers	EEC	2	0	0	2	100	-	100
MANDATORY COURSES									
2304GE551	Life skills V	MC	0	0	0	0	100	-	100
Total			20	0	8	24	630	470	1100

L – Lecture | T – Tutorial | P – Practical | C – Credit | CA - Continuous Assessment | ES - End Semester

2302IT501 OBJECT ORIENTED ANALYSIS AND DESIGN

L T P C
3 0 0 3

PREREQUISITE:

Software Engineering and Project Management

COURSE OBJECTIVES:

- 1.To Learn the basics of OO analysis and design skills.
- 2.To Learn the UML design diagrams.
- 3.To improve the software design with design patterns.
- 4.To be exposed to the various testing techniques.

COURSE OUTCOMES:

After completion of the course, Student will be able to

- CO1:** Design and implement projects using OO concepts
- CO2:** Use the UML analysis and design diagrams
- CO3:** Apply appropriate design patterns
- CO4:** Create code from design
- CO5:** Compare and contrast various testing techniques

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	-	2	3	-	-	-	-	-	3	-	-	1	-	-
CO2	-	-	3	3	-	-	-	-	-	-	-	-	-	2	-
CO3	1	-	3	3	-	-	-	-	-	2	-	-	-	-	1
CO4	-	2	-	-	-	3	3	-	-	2	-	-	-	1	-
CO5	-	-	-	-	2	3	3	-	-	1	-	-	-	-	1

COURSE CONTENTS:

MODULE I UML DIAGRAMS

9 Hours

Introduction to OOAD – Unified Process - UML diagrams – Use Case – Class Diagrams– Interaction Diagrams – State Diagrams – Activity Diagrams – Package, component and Deployment Diagrams

MODULE II DESIGN PATTERNS

9 Hours

GRASP: Designing objects with responsibilities – Creator – Information expert – Low Coupling – High Cohesion – Controller - Design Patterns – Creational - Structural -Behavioral.

MODULE III CASE STUDY

9 Hours

the Next Gen POS system, Inception -Use case Modeling - Relating Use cases - Elaboration - Domain Models - Finding conceptual classes and description classes – Associations – Attributes – Domain model refinement – Finding conceptual class Hierarchies - Aggregation and Composition

MODULE IV APPLYING DESIGN PATTERNS

9Hours

System sequence diagrams - Relationship between sequence diagrams and use cases Logical architecture and UML package diagram – Logical architecture refinement - UML class diagrams – UML interaction diagrams - Applying GoF design patterns

MODULE V TESTING

9 Hours

Object Oriented Methodologies — Software Quality Assurance — Mapping design to code – methodology – Survey – Rum Baugh, Booch, Jacobson Methods Testing: Issues in OO Testing – Class Testing – OO Integration Testing –GUI Testing – OO System Testing- Case Study : Vacation Tracking System

TOTAL: 45 HOURS

REFERENCES:

1. Craig Larman, "Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development", Third Edition, Pearson Education, 2016.
2. Simon Bennett, Steve Mc Robb and Ray Farmer, "Object Oriented Systems Analysis and Design Using UML", Fourth Edition, Mc-Graw Hill Education, 2012.

3. Erich Gamma, a n d Richard Helm, Ralph Johnson, John Vlissides, “Design patterns: Elements of Reusable Object-Oriented Software”, Addison-Wesley, 2012.
4. Martin Fowler, “UML Distilled: A Brief Guide to the Standard Object Modeling Language”, Third edition, Addison Wesley, 2013.
5. Paul C. Jorgensen, “Software Testing:- A Craftsman’s Approach”, Third Edition, Auerbach Publications, Taylor and Francis Group, 2010.
6. <http://nptel.ac.in/>

2302IT502

WEB PROGRAMMING

L T P C
3 0 0 3

PREREQUISITE:

Programming in Java Programming, Database Management Systems

COURSE OBJECTIVES:

1. Understand the scripting languages XHTML, JavaScript and PHP
2. Familiar with the different server technologies
3. Gain knowledge in the concepts of web services.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

- CO1:** Demonstrate the technologies used to create web pages
- CO2:** Design dynamic and interactive web pages by embedding Java Script in XHTML
- CO3:** Develop server side programming and apply database concepts to solve real time application
- CO4:** Build web applications using PHP and to represent data in XML format
- CO5:** Use AJAX and web services to develop interactive web applications

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	-	3	-	-	-	-	-	2	-	1	2	1
CO2	3	3	3	-	3	-	-	-	-	-	3	-	2	3	3
CO3	3	3	3	2	3	-	-	-	-	2	3	-	3	3	3
CO4	3	3	3	2	3	-	-	-	-	2	3	-	3	3	3
CO5	3	3	3	2	3	-	-	-	-	2	3	-	3	3	3

COURSE CONTENTS:

MODULE I WEB DESIGNING

9 Hours

Markup languages – Introduction to XHTML-Editing XHTML-Headings- Linking –Tables-Images-Forms- Internal linking – Frames – Lists
Cascading Style Sheets (CSS): Features-Style rule cascading and inheritance - Text properties –CSS box model

MODULE II CLIENT SIDE PROGRAMMING

9 Hours

Introduction to java script –Control statements I– Control statements II - Functions- Objects – Arrays- Introduction to JSON.

MODULE III SERVER SIDE PROGRAMMING

9 Hours

Servlets: Java Servlet Architecture- Servlet Life Cycle- Form GET and POST actions- Session Handling- Understanding Cookies- Installing and Configuring Apache Tomcat Web Server

DATABASE CONNECTIVITY: JDBC perspectives, JDBC program example — JSP: Understanding Java Server Pages-JSP Standard Tag Library (JSTL)-Creating HTML forms by embedding JSP code

MODULE IV PHP&XML

9Hours

An introduction to PHP: PHP- Using PHP- Variables- Program control- Built-in functions- Form Validation- Regular Expressions — File handling — Cookies — Connecting to Database.

XML: Basic XML- DTD - XML Schema, XML Parsers and Validation, XSLT Transformation

MODULE V INTRODUCTION AJAX AND WEB SERVICES

9 Hours

AJAX: Ajax Client Server Architecture-XML Http Request Response Object
Web Services: Introduction- Java web services Basics — Creating, Publishing, Testing and Describing a Web services (WSDL)-Consuming a web service-SOAP.

TOTAL: 45 HOURS

REFERENCES:

1. Paul Deitel, Harvey Deitel, Abbey Deitel, "Internet & World Wide Web - How to Program", 5th edition, Pearson

Education, 2012.

2.Robin Nixon, "Learning PHP, MySQL, JavaScript, CSS & HTML5" Sixth Edition, O'Reilly publishers, 2022.

3.Jeffrey C and Jackson, "Web Technologies A Computer Science Perspective", Pearson Education, 2011.

4.Steven Holzener , "PHP – The Complete Reference", 1st Edition, Mc-Graw Hill, 2017

5.Gopalan N.P. and Akilandeswari J., "Web Technology", Prentice Hall of India, 2011

6.UttamK.Roy, "Web Technologies", Oxford University Press, 2011

7.Learning Resources

<https://www.freecodecamp.org/>

<https://www.w3schools.com/>

<https://www.codecademy.com/>

2302IT503

MOBILE COMPUTING

L T P C
3 0 0 3

PREREQUISITE:

Principles of Communication, Computer Networks

COURSE OBJECTIVES:

- 1.To understand the basic concepts of mobile computing.
- 2.To learn the basics of mobile telecommunication system.
- 3.To be familiar with the network layer protocols and Ad-Hoc networks.
- 4.To know the basis of transport and application layer protocols.
- 5.To gain knowledge about different mobile platforms and application development.

COURSE OUTCOMES:

After completion of the course, Student will be able to

- CO1:** Explain the concepts of mobile computing schemes and applications
- CO2:** Illustrate the generations of telecommunication systems in wireless networks
- CO3:** Determine the functionality of MAC, network layer and Identify a routing protocol for a given Ad hoc network
- CO4:** Explain the functionality of Transport and Application layers
- CO5:** Develop a mobile application using android/blackberry/Ios/Windows SDK

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	2	-	2	-	2	2	1
CO2	3	2	-	-	-	-	-	-	2	-	3	-	3	1	1
CO3	3	3	2	2	-	-	-	-	-	-	2	-	3	1	2
CO4	3	2	-	2	-	-	-	-	-	-	2	-	3	1	2
CO5	3	3	3	3	3	2	-	2	2	3	3	-	2	3	3

COURSE CONTENTS:

MODULE I INTRODUCTION

9 Hours

Introduction to Mobile Computing – Applications of Mobile Computing- Generations of Mobile Communication Technologies- Multiplexing – Spread spectrum -MAC Protocols – SDMA- TDMA-CDMA

MODULE II MOBILE TELECOMMUNICATION SYSTEM

9 Hours

Introduction to Cellular Systems - GSM – Services & Architecture – Protocols – Connection Establishment –Frequency Allocation –Routing –Mobility Management – Security – GPRS-UMTS –Architecture – Handover – Security.

MODULE III MOBILE NETWORK LAYER

9 Hours

Mobile IP – DHCP – AdHoc– Proactive protocol-DSDV, Reactive Routing Protocols – DSR,AODV , Hybrid routing –ZRP, Multicast Routing- ODMRP, Vehicular Ad Hoc networks(VANET) – MANET Vs VANET – Security

MODULE IV MOBILE TRANSPORT AND APPLICATION LAYER

9Hours

Mobile TCP– WAP – Architecture – WDP – WTLS – WTP –WSP – WAE – WTA Architecture– WML.

MODULE V MOBILE PLATFORMS AND APPLICATIONS

9 Hours

Mobile Device Operating Systems – Special Constraints & Requirements – Commercial Mobile Operating Systems – Software Development Kit: iOS, Android, BlackBerry, Windows Phone – MCommerce – Structure –Pros & Cons –Mobile Payment System – Security Issues

TOTAL: 45 HOURS

REFERENCES:

- 1.Prasant Kumar Pattnaik, Rajib Mall, “Fundamentals of Mobile Computing”, PHI Learning Pvt. Ltd, New Delhi – 2016.
- 2.Jochen H. Schller, “Mobile Communications”, Second Edition, Pearson Education, New Delhi, 2014.

- 3.DharmaPrakashAgarval, Qing and An Zeng, "Introduction to Wireless and Mobile systems", Thomson Asia Pvt Ltd, 2013.
- 4.Uwe Hansmann, LotharMerk, Martin S. Nicklons and Thomas Stober, "Principles of Mobile Computing", Springer, 2012.
- 5.William.C.Y.Lee, "Mobile Cellular Telecommunications-Analog and Digital Systems", Second Edition, Tata McGraw Hill Edition ,2012.
- 6.C.K.Toh, "AdHoc Mobile Wireless Networks", First Edition, Pearson Education, 2012.
- 7.<http://nptel.ac.in/>

2302IT504 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

L T P C
3 0 2 4

PREREQUISITE:

- 1.Datamining
- 2.Programming for problem solving

COURSE OBJECTIVES:

- 1.Learn about the fundamentals of AI and ML.
- 2.Learn about the probabilistic reasoning.
- 3.Understand the Supervised learning with various techniques.
- 4.Study the extraction of k-means ,KNN, Gaussian models and Expectation maximization
- 5.Learn about Machine Learning based on various applications

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

- CO1:** Demonstrate the core concepts of Artificial Intelligence and Machine Learning with architecture.
- CO2:** Apply AI problem-Solving Techniques to implement common AI strategies using algorithms.
- CO3:** Implement Machine Learning Models (Supervised and Unsupervised).
- CO4:** Analyse and combining multiple learners using different techniques.
- CO5:** Understand the application of AI in areas like performance measurement and Image recognition,etc

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	-	3	-	-	-	-	-	3	-	3	3	2
CO2	3	2	2	-	2	-	-	-	-	-	2	-	3	2	2
CO3	3	3	3	-	3	-	-	-	-	-	3	-	3	3	2
CO4	3	3	3	-	3	-	-	-	-	-	3	-	3	3	2
CO5	3	2	2	-	2	-	-	-	-	-	2	-	3	2	2

COURSE CONTENTS:

MODULE I INTRODUCTION TO DATA WAREHOUSING

9 Hours

Introduction to AI – AI applications – problem solving agents – search algorithms – uninformed search strategies – Heuristic search strategies – Local search and optimization problems – advanced search – constraint satisfaction problems (CSP).

MODULE II PROBABILISTIC REASONING

9 Hours

Acting under uncertainty – Bayesian inference - naïve Bayesian models, Probabilistic reasoning - Bayesian networks – exact inference in BN – approximate inference in BN – casual networks

MODULE III SUPERVISED LEARNING

9 Hours

Introduction to machine learning – Linear Regression Models: Least squares, single & multiple variables, Bayesian linear regression, gradient descent, Linear Classification Models: Discriminant function – Probabilistic discriminative model - Logistic regression, Probabilistic generative model – Naive Bayes, Maximum margin classifier – Support vector machine, Decision Tree, Random forests

MODULE IV ENSEMBLE TECHNIQUES AND UNSUPERVISED LEARNING

9Hours

Combining multiple learners: Model combinations schemes, Voting, Ensemble Learning – bagging, Boosting, stacking, Unsupervised learning: k – means, Instance based Learning: KNN, Gaussian mixture models and Expectation maximization.

MODULE V MACHINE LEARNING IN PRACTICE AND APPLICATIONS

9 Hours

Perceptron - Multilayer perceptron, activation functions, network training – gradient descent optimization – stochastic gradient descent, error back propagation, from shallow networks to deep networks –Unit saturation (aka the vanishing gradient problem) – ReLU, hyper parameter tuning, batch normalization,

regularization, dropout.

PRACTICAL EXERCISES:

1. Implementation of Uninformed search algorithms (BFS, DFS)
2. Implementation of Informed search algorithms (A*, memory-bounded A*)
3. Write a program to implement the naïve Bayesian classifier for credit card analysis and compute the accuracy with a few test data sets
4. Build Regression models
5. Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.
6. Build SVM models
7. Implement Ensembling techniques
8. Implement clustering algorithms
9. Implement EM for Bayesian networks
10. Build deep learning NN models

List of Equipments:(30 Students per Batch)

Tools: Python, Numpy, Scipy, Matplotlib, Pandas, statmodels, seaborn, plotly, bokeh

Note: Example data sets like: UCI, Iris, Pima Indians Diabetes etc

TOTAL: 45 HOURS

REFERENCES:

1. *Stuart Russell and Peter Norvig, "Artificial Intelligence – A modern Approach", Fourth Edition , Pearson Education, 2021.*
2. *Ethm Alpaydin, " Introduction to Machine Learning ", MIT Press , Fourth Edition, 2020*
3. *Ameet V Joshi, —Machine Learning and Artificial Intelligence, Springer Publications, 2020.*
4. *Dan W. Patterson, "Introduction to Artificial Intelligence and Expert Systems", Pearson Education, 2007*
6. <http://nptel.ac.in/>
7. <http://www.simpilearn.com/>
8. <http://towardsdatascience.com/>

2303IT003

BIG DATA

L T P C
3 0 0 3

PREREQUISITE:

Java Programming , Database Management Systems & Basic Linux Commands

COURSE OBJECTIVES:

1. To understand the need of a framework to store and process the big data.
2. To have knowledge on the Big Data Technologies for processing the Different types of Data.
3. To understand the advanced frame work for faster accessing and processing of Big Data.

COURSE OUTCOMES:

After completion of the course, Student will be able to

- CO1:** Understand the need for new frameworks to handle massive data efficiently.
- CO2:** Perform data operations using NoSQL databases like HBase, Cassandra, and MongoDB in real-world applications.
- CO3:** Analyze the architecture and components of the Hadoop ecosystem for distributed data storage and processing.
- CO4:** Automate data workflows using Pig and Hive for efficient management.
- CO5:** Implement Spark programming with various languages and apply graph algorithms for data processing.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	1	-	-	-	-	-	-	2	-	2	1	1
CO2	3	2	2	2	3	-	-	-	-	-	3	-	3	3	3
CO3	3	3	2	2	2	-	-	-	-	-	3	-	3	2	2
CO4	3	3	3	2	3	-	-	-	-	-	3	-	3	3	2
CO5	3	3	3	3	3	-	-	-	-	-	3	-	3	3	2

COURSE CONTENTS:

MODULE I UNDERSTANDING BIG DATA

9 Hours

Introduction to Big Data - Concepts and terminology, Big Data Characteristics, Different types of Data -Big Data Challenges - Big Data Architecture-- Big Data Storage: File system and Distributed File System-- Big Data applications-- Big Data technologies- Cloud and Big Data

MODULE II DATA STORAGE AND MANAGEMENT

9 Hours

Introduction to NoSQL – aggregate data models - Hbase – data model and implementations – Hbase clients – Hbase examples- Cassandra – Cassandra data model – Cassandra examples – Cassandra clients- MongoDB – Architecture – Schema Design and Modelling – CRUD operations

MODULE III HADOOP FRAMEWORK

9 Hours

Hadoop Architecture - Hadoop Distributed File System (HDFS) –HDFS Concepts-YARN – Hadoop I/O – Map Reduce: Developing a map-reduce application- Hadoop streaming – Hadoop pipes- Java interface – data flow- serialization – Avro – file-based data structures

MODULE IV ESSENTIAL TOOLS OF BIG DATA

9Hours

Introduction to Pig- Pig Architecture, Grunt, Pig Data Model-scalar and complex types. Pig Latin- Input and output, Relational operators, User defined functions -Working with scripts-Introduction to Hive- UNITS, Data types and file formats, Hive QL queries, Hive QL views- reduce query complexity. Hive scripts-Hive QL Indexes- Aggregate functions- Bucketing vs Partitioning.

MODULE V BIG DATA PROCESSING WITH APACHE SPARK

9 Hours

Overview of Spark -Cluster Design and Management – Application Programming interface (API): Spark Context, Resilient Distributed Datasets- Spark Programming in Scala, Python, R, Java - Spark SQL: SQL Context – Importing and Saving data – Data frames – using SQL – GraphX: overview – Creating Graph – Graph Algorithms

TOTAL: 45 HOURS

REFERENCES:

1. Thomas Erl, Wajid Khattak, and Paul Buhler, *Big Data Fundamentals: Concepts, Drivers & Techniques*, Pearson India Education Service Pvt. Ltd., First Edition, 2016.
2. Tom White, "*Hadoop: The Definitive Guide*", O'Reilly, 4th Edition, 2015.
3. Alan Gates, *Programming Pig Dataflow Scripting with Hadoop*, O'Reilly Media, Inc, 2011.
4. Jason Rutherglen, Dean Wampler, Edward Capriolo, *Programming Hive*, O'Reilly Media Inc, 2012
5. Mike Frampton, "*Mastering Apache Spark*", Packt Publishing, 2015.
6. <https://www.oracle.com/in/big-data/what-is-big-data/>
7. <https://www.coursera.org/courses?query=big%20data>

2303IT014

DEVOPS

L	T	P	C
3	0	0	3

PREREQUISITE:

Operating System , Software Engineering

COURSE OBJECTIVES:

- 1.To provide an in-depth understanding of DevOps tools and their role in fostering automation and collaboration in team environments.
- 2.To impart practical knowledge on applying version control techniques for managing complex, large-scale software projects.
- 3.To equip students with the skills to use Jenkins for designing and automating CI/CD pipelines, enhancing build and deployment efficiency
- 4.To develop expertise in ensuring system resilience and scalability through effective monitoring, scaling, and configuration management practices.
- 5.To integrate security principles into DevOps workflows, enabling secure, robust, and efficient application development and deployment processes

COURSE OUTCOMES:

After completion of the course, Student will be able to

- CO1:** Understand the role of DevOps tools in automation and team collaboration.
- CO2:** Apply version control best practices for efficient management of large-scale software projects.
- CO3:** Use Jenkins to automate CI/CD pipelines for building and deploying applications.
- CO4:** Ensure system reliability with monitoring, scaling, and configuration management tools.
- CO5:** Integrate security into DevOps workflows for secure and efficient application development.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	1	1	3	-	-	3	2	2	3	-	1	2	2
CO2	3	2	2	2	3	-	-	3	2	3	3	-	1	3	2
CO3	3	3	3	2	3	-	-	2	2	3	3	-	1	3	3
CO4	3	3	3	3	3	-	-	2	2	3	3	-	1	3	3
CO5	3	3	3	2	3	2	3	2	2	3	3	-	1	3	3

COURSE CONTENTS:

MODULE I INTRODUCTION TO DEVOPS

9 Hours

Introduction to DevOps-DevOps Principles-Skills of a DevOps Engineer-DevOps Lifecycle-DevOps Delivery Pipeline-Market Trends and Challenges in DevOps-Overview of DevOps Tools

MODULE II VERSION CONTROL WITH GIT

9 Hours

Introduction to version control-Basics of Git-Installing and configuring Git-Common Git commands

-Branching and merging strategies-Working with remote repositories-Collaborating with Teams-Managing Conflicts-Overview of Git Hosting Platforms: GitHub, GitLab, Bitbucket.

MODULE III CONTINUOUS INTEGRATION AND CONTINUOUS DEPLOYMENT(CI/CD)

9 Hours

Introduction to CI/CD Concepts- CI/CD Process-Jenkins Architecture and Setup-Building and Deploying Applications Using Jenkins-Pipeline as code with Jenkins -Advanced CI/CD Techniques

MODULE IV CONTAINERIZATION, ORCHESTRATION & CONFIGURATION MANAGEMENT

9Hours

Introduction to Containerization with Docker-Working with Docker-Advanced Docker Features-Introduction to Kubernetes - Kubernetes architecture and components-Setting up a Kubernetes cluster-Introduction to Ansible - Ansible architecture and installation-Inventory management-Ad-hoc commands and playbooks

MODULE V CONTINUOUS MONITORING AND LOGGING, CLOUD COMPUTING & DEVSECOPS

9 Hours

Introduction to Nagios and Prometheus-Setting up monitoring with Nagios-Monitoring applications with Prometheus-DevOps on AWS, Azure, and Google Cloud-Setting up CI/CD pipelines in the cloud-Cloud-native DevOps practices- Introduction to DevSecOps-Securing the CI/CD pipeline-Container security best practices.

TOTAL: 45 HOURS

REFERENCES:

- 1.*The DevOps Handbook (2nd Edition)*by Gene Kim, Jez Humble, Patrick Debois, John Willis, and Nicole Forsgren
- 2.*Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation (1st Edition)*by Jez Humble and David Farley
- 3.*Docker Deep Dive (4th Edition)*by Nigel Poulton
- 4.*"Kubernetes: Up and Running"*by Brendan Burns, Joe Beda, and Kelsey Hightower
- 5.RobertoVormittag, *"A Practical Guide to Git and GitHub for Windows Users: From Beginner to Expert in Easy Step-By-Step Exercises"*, Second Edition, Kindle Edition, 2016.
- 6.Jeff Geerling, *"Ansible for DevOps: Server and configuration management for humans"*, First Edition, 2015
- 7.Len Bass, Ingo Weber and Liming Zhu, *"DevOps: A Software Architect's Perspective"*, Pearson Education, 2016
- 8.<https://www.jenkins.io/user-handbook.pdf>

2303IT037	DATA WAREHOUSING AND DATA MINING	L	T	P	C
		3	0	0	3

PREREQUISITE:

- 1.Database Management Systems
- 2.Java Programming
- 3.Python Programming

COURSE OBJECTIVES:

- 1.Learn about the safe storage of data and architecture of data warehouse.
- 2.Learn about the Elimination of errors from the data.
- 3.Understand the Deleting data that is no longer important to the organization.
- 4.Study the extraction of implicit, previously unknown, and potentially useful information from data
- 5.To help in the generation of reports for the management.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

- CO1:** Explain the concepts of Data Warehousing Architecture and Implementations.
CO2: Apply different association rules to solve various dataset.
CO3: Design high dimensional data analytics system using classification and prediction techniques.
CO4: Analyse large dataset using clustering and association techniques.
CO5: Analyse various complex data objects and real time dataset using advanced mining techniques.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	-	3	-	-	-	-	-	3	-	3	3	2
CO2	3	2	2	-	2	-	-	-	-	-	2	-	3	2	2
CO3	3	3	3	-	3	-	-	-	-	-	3	-	3	3	2
CO4	3	3	3	-	3	-	-	-	-	-	3	-	3	3	2
CO5	3	2	2	-	2	-	-	-	-	-	2	-	3	2	2

COURSE CONTENTS:

MODULE I INTRODUCTION TO DATA WAREHOUSING

9 Hours

Overview and Concepts: Data Ware House Components - Building a Data Warehouse - Data Ware House Architecture - Online Analytical Processing(OLAP)- Multi Dimensional Data Analysis

MODULE II DATA MINING PRIMITIVES

9 Hours

Introduction to Data mining – Knowledge Discovery in Database(KDD)- Preprocessing – Predictive Analysis - Data mining primitives – Data mining query language

MODULE III DATA MINING TECHNIQUES

9 Hours

Introduction to Classification– Decision Tree Induction – Bayesian Classification – Back propagation – Lazy Learners – Prediction

MODULE IV CLUSTERING AND ASSOCIATION

9Hours

Hierarchical Algorithms – Partition Algorithms – Outlier Analysis – Mining Frequent Patterns, Associations, and Correlations

MODULE V ADVANCED TOPICS

9 Hours

Web Mining – Web Content Mining – Spatial Mining – Time Series and Sequence Mining – Graph Mining.

TOTAL: 45 HOURS

REFERENCES:

1. Jiawei. Han, *Design high dimensional data analytics system using classification and prediction techniques*. Micheline Kamber, “*Data Mining: Concepts and Techniques*”, Second Edition, Elsevier, New Delhi, 2017
2. Vipin Kumar, Michael Steinbach, “*Introduction to Data Mining*”, Second Edition, Addison Wesley, 2015
3. Dunham M, —*Data Mining: Introductory and Advanced Topics*], Prentice Hall, New Delhi, 2013.
4. <http://nptel.ac.in/>

2303IT038 MULTIMEDIA SYSTEMS AND ITS APPLICATIONS

L T P C
3 0 0 3

PREREQUISITE:

Computer Fundamentals and Computer Graphics Basics

COURSE OBJECTIVES:

- 1.Understand the fundamental concepts and components of multimedia systems.
- 2.To learn representations, perceptions and application of Multimedia
- 3.Explain multimedia text, image, and graphics representation
- 4.Explore computer animation and digital video fundamentals
- 5.Evaluate and implement multimedia solutions

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

- CO1:** Explain the fundamental concepts and classifications of multimedia.
CO2: Describe and apply audio technologies and file formats used in multimedia
CO3: Demonstrate understanding of text, image, and graphics handling in multimedia
CO4: Analyse and implement basic principles of animation and digital video production.
CO5: Design and develop a multimedia project

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	-	-	-	-	-	1	-	1	-	3	-	1
CO2	3	3	2	-	-	-	-	-	-	-	-	-	3	1	1
CO3	3	2	1	-	-	-	-	-	-	-	-	-	3	2	1
CO4	3	2	2	1	2	-	-	-	-	2	-	-	3	3	1
CO5	3	2	2	-	2	-	-	-	3	2	2	-	3	3	1

COURSE CONTENTS:

MODULE I INTRODUCTION TO MULTIMEDIA

9 Hours

Definition-Classification - Multimedia Application - Multimedia Hardware - Multimedia Software - Multimedia Systems Architecture - CDROM – DVD.

MODULE II MULTIMEDIA DATA REPRESENTATION

9 Hours

Multimedia Audio: Digital Medium – Digital Audio Technology – Sounds Cards – Recording – Editing – MIDI Fundamentals – Working With MIDI – Audio File Formats – Adding Sound To Multimedia Project

MODULE III MULTIMEDIA COMPRESSION TECHNIQUES

9 Hours

Multimedia Text: Text in Multimedia – Multimedia Graphics: Coloring – Digital Imaging Fundamentals – Development and Editing – File Formats – Scanning and Digital Photography.

MODULE IV MULTIMEDIA ANIMATION AND VIDEO

9Hours

Computer Animation Fundamentals – Kinematics – Morphing – Animation S/W tools and Techniques – Video Works – Digital Video Fundamentals – Digital Vide Production and Editing Techniques.

MODULE V MULTIMEDIA PROJECT AND AUTHORING

9 Hours

Multimedia Stages Of Project – Multimedia Skills - Design Concept – Authoring – Planning and Costing – Multimedia Team – Futures – Digital Communication and New Media, Interactive Television, Digital Broad Casting Digital Radio, multimedia Conferencing

TOTAL: 45 HOURS

REFERENCES:

- 1.S.Gokul "Multimedia Magic",BPB Publication,2nd Edition.
- 2.TayVaughen,"Multimedia making it Work",TMH,6th Edition.
- 3.Ze-Nian Li and Mark S. Drew, "Fundamentals of Multimedia", 2nd Edition, Springer, 2014.
- 4.Prabhat K. Andleigh and KiranThakrar, "Multimedia Systems Design", Pearson Education, 2003.
- 5.Nigel Chapman and Jenny Chapman, "Digital Multimedia", 2nd Edition, Wiley India, 2009.
- 6.<http://nptel.ac.in/>

2302IT551

WEB PROGRAMMING LABORATORY

L	T	P	C
0	0	2	1

PREREQUISITE:

Programming in Java Programming, Database Management Systems

COURSE OBJECTIVES:

- 1.To develop an ability to design and implement static and dynamic website
- 2.Use appropriate client-side or Server-side applications
- 3.Understand, analyze and create XML documents and XML Schema

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

- CO1:** Create web pages using HTML and Cascading Styles sheets
- CO2:** Create dynamic web pages using JavaScript
- CO3:** Develop Servlet and JSP applications to implement Session management and Data base Connectivity
- CO4:** Build web applications using PHP
- CO5:** Design and validate XML documents.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	3	-	3	-	-	-	-	-	1	-	1	2	2
CO2	3	2	3	-	3	-	-	-	-	-	2	-	2	3	3
CO3	3	3	3	2	3	-	-	-	-	2	3	-	3	3	3
CO4	3	3	3	2	3	-	-	-	-	2	3	-	3	3	3
CO5	3	2	2	-	3	-	-	-	-	-	2	-	3	2	3
CO6	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

LIST OF EXPERIMENTS:

Experiment 1 (HTML Page)

- Design the following static web pages required for an online book store web site.
- HOME PAGE: The static home page must contain three frames.
- LOGIN PAGE
- CATALOGUE PAGE: The catalogue page should contain the details of all the books available in the web site in a table.
- REGISTRATION PAGE

Experiment 2 (CSS)

- Develop and demonstrate the usage of inline, internal and external style sheet using CSS

Experiment 3 (JavaScript)

Write JavaScript to validate the following fields of the Registration page.

- First Name (Name should contains alphabets and the length should not be less than 6 characters).
- Password (Password should not be less than 6 characters length).
- E-mail id (should not contain any invalid and must follow the standard pattern name@domain.com)
- Mobile Number (Phone number should contain 10 digits only).
- Last Name and Address (should not be Empty).

Experiment 4 (Installing Servers)

- Install TOMCAT web server and APACHE. While installation assign port number 4040 to TOMCAT and 8080 to APACHE. Make sure that these ports are available i.e., no other process is using this port.
- Access the following static web pages - an online book store web site using these servers
- Access the pages by using the urls : <http://localhost:4040/rama/books.html> (for tomcat)
<http://localhost:8080/books.html> (for Apache)

Experiment 5 (Installing Databases)

- Install a database (Mysql or Oracle).
- Create a table which should contain at least the following fields: name, password, email-id, phone number (these should hold the data from the registration form).

- (c) Practice 'JDBC' connectivity.
- (d) Write a java program/servlet/JSP to connect to that database and extract data from the tables and display them.
- (e) Experiment with various SQL queries. Insert the details of the users who register with the web site, whenever a new user clicks the submit button in the registration page.

Experiment 6, 7 & 8 (Servlet, JSP and PHP)

- (a) A web application that takes a name as input and on submit it shows a hello <name> page where name is taken from the request.
- (b) It shows the start time at the right top corner of the page and provides a logout button.
- (c) On clicking this button, it should show a logout page with Thank You <name > message with the duration of usage (hint: Use session to store name and time).

Experiment 9 (XML with DOM)

- (a) Create an XML document that contains 10 users information.
Write a Java Program, which takes User Id as input and returns the user details by taking the user information from XML document using DOM parser or SAX parser.

TOTAL: 30 HOURS

REFERENCES:

1. Paul Deitel, Harvey Deitel, Abbey Deitel, "Internet & World Wide Web - How to Program", 5th edition, Pearson Education, 2012.
2. Robin Nixon, "Learning PHP, MySQL, JavaScript, CSS & HTML5" Third Edition, O'Reilly publishers, 2014.
3. Jeffrey C and Jackson, "Web Technologies A Computer Science Perspective", Pearson Education, 2011.
4. Steven Holzener, "PHP – The Complete Reference", 1st Edition, Mc-Graw Hill, 2017
5. Gopalan N.P. and Akilandeswari J., "Web Technology", Prentice Hall of India, 2011
6. Uttam K. Roy, "Web Technologies", Oxford University Press, 2011.
7. Learning Resources
<https://www.freecodecamp.org/>
<https://www.w3schools.com/>
<https://www.codecademy.com/>

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CASE TOOLS LAB (MINI PROJECT 1)

L	T	P	C
0	0	2	1

PREREQUISITE:

Software Engineering and Project Management

COURSE OBJECTIVES:

- 1.Understand the concept of modeling and mechanism involved in UML.
- 2.Learn the classes and different types of relationships in classes, objects and terms related to diagrams.
- 3.Examine fundamental object-oriented analysis and design techniques.
- 4.Apply design patterns for viewing a system as a set of procedures.
- 5.Prepare case studies for analyzing modeling techniques.

COURSE OUTCOMES:

After completion of the course, Student will be able to

- CO1:** Design and implement projects using OO concepts.
- CO2:** Use the UML analysis and design diagrams.
- CO3:** Apply appropriate design patterns.
- CO4:** Create code from design.
- CO5:** Compare and contrast various testing techniques.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	3	-	-	2	2	2	3	-	1	3	3
CO2	2	3	3	2	3	-	-	-	3	-	2	-	1	2	3
CO3	3	3	3	1	2	-	-	-	-	-	3	-	1	3	3
CO4	3	2	3	2	3	-	-	-	-	-	2	-	1	3	3
CO5	3	3	1	3	2	-	-	-	-	-	3	-	1	2	3
CO6	1	1	2	1	3	-	-	-	-	-	-	-	1	1.8	1.2

TO DEVELOP A MINI-PROJECT USING FOLLOWING PROBLEM STATEMENTS

Identify Use Cases and develop the Use Case model.

- Identify the conceptual classes and develop a domain model with UML Class diagram.
- Using the identified scenarios, find the interaction between objects and represent them using UML Sequence diagrams.
- Draw relevant state charts and activity diagrams.
- Test the software system for all the scenarios identified as per the usecase diagram
- Identify the User Interface, Domain objects, and Technical services. Draw the partial layered, logical architecture diagram with UML package diagram notation.
- Implement the modified system and test it for various scenarios.
- Develop and test the Technical services layer.
- Develop and test the Domain objects layer.
- Develop and test the User interface layer

Domains for Mini Project

- Online Purchase System
- Library Management System
- E-Ticketing
- Quiz System
- Student Mark Analyzing System
- E-Mail Client System
- Telephone Phone Dialing
- Point of Sale
- Software personnel management system

Airline/Railway reservation system

Requirements: Argo UML or Eclipse IDE or Rational Suite or Visual Paradigm or equivalent

TOTAL: 30 HOURS

REFERENCES:

1. Bernd Bruegge and Allen H. Dutoit, “Object-Oriented Software Engineering: Using UML, Patterns and Java”, Third Edition, Pearson Education, 2009.
2. Roger S. Pressman, Object-Oriented Software Engineering: An Agile Unified Methodology, First Edition, McGraw-Hill International Edition, 2014.
3. Craig Larman, Applying UML and Patterns, 3rd ed, Pearson Education, 2005.

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STARTUP OPPORTUNITIES FOR IT ENGINEERS

L T P C
2 0 0 2

PREREQUISITE:

1. Basic understanding of Entrepreneurship
2. Problem solving skills
3. Business Strategy Management
4. Technology fundamentals

COURSE OBJECTIVES:

1. Learn how to harness technology for groundbreaking entrepreneurial ventures.
2. Explore creative ways to identify and seize opportunities.
3. Learn to gather data and analyze trends to ensure your business concept is viable.
4. Develop skills to turn ideas into actionable business propositions and deliver impactful presentations.
5. Understand the fundamentals of launching new ventures and building cohesive teams.
6. Develop the ability to translate visionary ideas into detailed plans for market execution and financial viability.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

- CO1:** Explain Concepts of Innovation, Entrepreneurship and Startups in Technology
- CO2:** Develop Startup or Business ideas, minimum viable products and business models for real life problems
- CO3:** Analyze global trends in technology and entrepreneurship to identify opportunities for innovation and competitive advantage.
- CO4:** Apply market research techniques and conduct feasibility analyses to identify opportunities for innovative technological solutions.
- CO5:** Demonstrate teamwork and communication skills in multidisciplinary teams to pitch innovative startup ideas effectively.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	-	-	-	-	-	1	2	2	3	-	1	1	1
CO2	2	3	3	2	2	2	-	2	2	3	3	-	1	3	2
CO3	-	3	-	3	-	2	-	1	2	2	3	-	1	2	1
CO4	-	3	2	3	2	2	-	1	2	3	3	-	1	2	1
CO5	-	1	-	-	-	-	-	3	3	2	2	-	1	2	1
CO6	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

COURSE CONTENTS:

1. Introduction to Startups
2. Foundations of Entrepreneurship – Strategies & Competencies
3. Startup Feasibility Analysis, Survival & Growth
4. Ethics and Social Responsibilities in Startups
5. Startup Success Metrics & Business Plan
6. Customer – Centric Innovation
7. Leveraging Technologies for Startup Success
8. Managing New Venture and Secure Funding
9. Business Models and Value Proposition
10. Open Source in Startups
11. Digital Marketing and Strategies
12. Minimum Viable Product Development & Tools for Creating MVPs
13. Patent and Intellectual Property

14. Effective Team Building
15. Entrepreneurial Ecosystem Dynamics

Evaluation Procedure

1. Assignment 1– Review of IT based Startups – 20 Marks
2. Assignment 2– Open Source in Startups & Business Models – 30 Marks
3. Assignment 3 – Review of Business Models of IT based Startups – 20 Marks
4. Presentation – Idea, Market Research Results, 1st MVP, Possible Fund Raising Model to be adopted, IP – 30 Marks

TOTAL: 30 HOURS

REFERENCES:

1. *Zero to One: Notes on Startups, or How to Build the Future* by Peter Thiel and Blake Masters
2. *Start with Why: How Great Leaders Inspire Everyone to Take Action* by Simon Sinek
3. *The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses* by Eric Ries
4. <https://www.edx.org/course/innovation-and-entrepreneurship>.

